

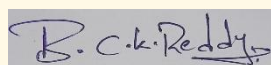
SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Scenario-Based

Course Code:	ITA0512	Course Title:	Computer Vision
CO Assessed:	CO2 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Scenario-Based
Weightage:	10% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)</i>		
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Batch	2024	Date:	08/08/2026

Code of Conduct

I B. Charan Kumar Reddy (192472195) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



RUBRICS FOR CALCULATION: ASSESSMENT TOOL 2: Scenario-Based

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

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Course Code: ITA 0512

Course Name : Computer Vision

Answers

26. Frequency-Based Enhancement

(a) Interpretation of the Scenario and Explanation of Frequency-Based Processing

Scenario Interpretation

The given scenario states that an image requires enhancement using **frequency-domain methods**. This means that the image quality needs to be improved by manipulating its frequency components rather than directly modifying individual pixels.

Image enhancement is generally performed to improve **contrast, sharpness, visibility of fine details, and overall image quality**. In the frequency domain, an image is represented in terms of its spatial frequency components.

- **Low-frequency components** represent slowly changing intensity regions, such as smooth backgrounds and gradual illumination variations.
- **High-frequency components** represent rapid intensity changes, such as edges, fine details, textures, and noise.

Therefore, frequency-based enhancement allows specific frequency components to be strengthened or suppressed according to the requirements of the image.

Frequency-Domain Processing

Frequency-domain processing involves the following steps:

Input Image → Fourier Transform → Frequency-Domain Filtering → Inverse Fourier Transform → Enhanced Image

First, the input image is converted from the spatial domain to the frequency domain using the **Discrete Fourier Transform (DFT)**. A frequency-domain filter is then applied to modify selected frequency components. Finally, the **Inverse Fourier Transform (IDFT)** converts the processed image back to the spatial domain.

The basic frequency-domain operation is:

$$G(u, v) = H(u, v)F(u, v)$$

Where:

- $F(u, v)$ = Fourier transform of the original image
- $H(u, v)$ = Frequency-domain filter
- $G(u, v)$ = Filtered frequency-domain image

The enhanced spatial image is obtained using:

$$g(x, y) = F^{-1}\{G(u, v)\}$$

Thus, frequency-domain processing provides direct control over different frequency components of an image.

(b) Identification of Key Issues in Spatial-Domain Methods

Spatial-domain methods directly modify the intensity values of individual pixels. Although they are simple and useful, they have some limitations when selective enhancement of image details is required.

1. Limited Frequency Selectivity

Spatial-domain operations work directly on pixel values and do not explicitly separate low- and high-frequency information.

For example, a simple sharpening operation may enhance both desired edges and unwanted noise because both contain high-frequency components.

2. Noise Amplification

Sharpening techniques in the spatial domain can increase noise along with useful image details.

For example, applying a strong sharpening mask to a noisy image may produce unwanted bright and dark points around objects.

3. Difficulty in Controlling Different Frequency Components

It is difficult to selectively control particular frequency ranges using simple spatial filters. Frequency-domain filtering provides direct control over low, middle, and high frequencies.

4. Edge and Detail Trade-Off

Smoothing can remove noise but may also blur important edges and fine details. On the other hand, sharpening may restore details but can amplify noise.

5. Global Illumination Problems

Spatial-domain enhancement may not effectively separate gradual illumination variations from important image details. Frequency-domain techniques can selectively suppress low-frequency variations when required.

Key Observation

Therefore, when the objective is to **selectively enhance fine details and edges**, frequency-domain processing is more suitable because frequency components can be directly analyzed and controlled.

(c) Application of an Appropriate Frequency-Domain Technique

Selected Technique: Gaussian High-Pass Filtering

For the given scenario, a **Gaussian High-Pass Filter (GHPF)** is selected because the objective is image enhancement and improvement of fine details.

A high-pass filter suppresses low-frequency components and allows high-frequency components to pass.

Since high-frequency components contain **edges, fine details, and textures**, enhancing them can make important image structures more visible.

Gaussian High-Pass Filter

The Gaussian low-pass filter is:

$$H_{LP}(u, v) = e^{-\frac{D^2(u, v)}{2D_0^2}}$$

Therefore, the Gaussian high-pass filter is:

$$H_{HP}(u, v) = 1 - e^{-\frac{D^2(u, v)}{2D_0^2}}$$

Where:

- $D(u, v)$ = Distance of the frequency point from the center of the frequency spectrum
- D_0 = Cut-off frequency

- $H_{HP}(u, v)$ = Gaussian high-pass filter

Application Procedure

Step 1: Read the input image

Let the input image be represented as:

$$f(x, y)$$

Step 2: Convert the image to frequency domain

Apply the Fourier Transform:

$$F(u, v) = DFT\{f(x, y)\}$$

Step 3: Construct the Gaussian high-pass filter

$$H_{HP}(u, v) = 1 - e^{-\frac{D^2(u, v)}{2D_0^2}}$$

Step 4: Apply the filter

Multiply the Fourier spectrum by the filter:

$$G(u, v) = H_{HP}(u, v)F(u, v)$$

Step 5: Convert back to spatial domain

Apply the inverse Fourier transform:

$$g(x, y) = IDFT\{G(u, v)\}$$

Step 6: Obtain the enhanced image

The resulting image contains enhanced high-frequency information, making edges and fine details more visible.

Simple Representation of the Process

Input Image $\rightarrow$ *DFT* $\rightarrow$ *Gaussian High – Pass Filter* $\rightarrow$ *IDFT* $\rightarrow$ *Enhanced Image*

(d) Logical Justification of the Choice

The **Gaussian High-Pass Filter** is selected because the scenario specifically requires image enhancement using frequency-domain processing.

The choice is justified for the following reasons:

1. Enhances Fine Details

High-frequency components correspond to rapid changes in intensity, including edges and fine image details. Passing these components makes important structures more prominent.

2. Provides Frequency-Based Control

Unlike simple spatial-domain operations, the frequency-domain filter allows direct control over which frequency components are enhanced or suppressed.

3. Gaussian Response Produces Smooth Filtering

A Gaussian high-pass filter provides a gradual transition between suppressed and retained frequencies. This avoids the abrupt frequency cutoff associated with an ideal high-pass filter.

4. Reduces Abrupt Artifacts

Ideal frequency filters can introduce ringing artifacts because of their sharp cutoff. The Gaussian response changes smoothly, making it more suitable when smooth enhancement is required.

5. Suitable for Image Enhancement

The technique is useful when the image contains important edges, textures, and fine structures that need to be made more visible.

Example 1: Medical Image

In an X-ray or medical image, fine boundaries between structures may be difficult to observe. High-frequency enhancement can make these boundaries more visible.

Example 2: Satellite Image

In a satellite image, roads, building boundaries, and other fine structures can be enhanced using high-frequency components.

Comparison of Spatial and Frequency-Domain Enhancement

Feature	Spatial-Domain Method	Frequency-Domain Method
Processing	Directly modifies pixels	Modifies frequency components
Frequency control	Limited	Direct and selective
Edge enhancement	Possible	Highly controllable
Noise control	May be difficult	Frequency components can be selectively suppressed
Fine-detail enhancement	Possible	Effective
Mathematical basis	Spatial masks/operators	Fourier Transform and frequency filters
Suitable for	Simple local enhancement	Selective frequency-based enhancement

(e) Clear Explanation and Final Conclusion

Overall Decision

For the given scenario, **frequency-domain enhancement using a Gaussian High-Pass Filter is an appropriate choice.**

The image is first transformed into the frequency domain using the Fourier Transform. The Gaussian high-pass filter then suppresses low-frequency components while allowing high-frequency components containing edges and fine details to pass. The filtered frequency spectrum is finally transformed back into the spatial domain using the inverse Fourier transform. The technique provides better control over image frequencies than simple spatial-domain methods and can improve the visibility of important image details.

Conclusion

Frequency-based image enhancement is useful when specific frequency components of an image need to be controlled. Spatial-domain methods directly manipulate pixel intensities but may have limitations in selective frequency control and may amplify noise during sharpening. Therefore, a **Gaussian High-Pass Filter** is selected for the given scenario because it enhances edges and fine details while providing a smooth frequency response.

Input Image $\rightarrow$ DFT $\rightarrow$ Gaussian High – Pass Filter $\rightarrow$ IDFT $\rightarrow$ Enhanced Image

Final Decision:

Gaussian High-Pass Filtering is the most appropriate technique for enhancing fine details and edges in the given frequency-domain enhancement scenario.